Full bench analysis report

Speaker playback, microphone recording, timing and frequency response

SONIC Auditory Benchmarking Data acquired 23 October 2026

1 Findings and decision summary

The wired playback and microphone-recording path is operational. Analysis of four saved native captures establishes a repeatable *application sample-grid waveform roundtrip* of approximately **38.3 ms** with nonperiodic timing probes. Both SONIC recordings contain components at every requested Sheep E frequency, but their recorded levels are substantially unequal. These findings support a working bench demonstration; they do not yet qualify calibrated delivery across the full SONIC alphabet.

Question	Finding from the saved data
Playback and recording together?	Verified by the known-melody check and the two recorded SONIC blocks.
Measured roundtrip?	Forty of forty sweeps accepted. Pooled median 38.307 ms; observed range 38.240–38.375 ms.
Repeatability?	Two stream-start medians: 38.250 and 38.375 ms. Their difference is 0.125 ms.
All 14 SONIC components?	Observed in both microphone recordings, with control and neighboring-tone checks. This is recorded pickup, not isolated acoustic qualification.
Uniform frequency level?	Not established. Relative to 2 kHz, 17–19.5 kHz was about 15–22 dB lower; 20.9–25.8 kHz about 30–40 dB lower.
SPL, distortion, acoustic onset?	Unqualified: microphone correction, absolute level calibration, acoustic-isolation controls and independent physical timing were not available.

Table 1: Main conclusions. Numeric delays refer to the sweep-based application roundtrip, not a delay measured directly from SONIC tones.

Physical chain. Computer/Core Audio → Audient iD14 → wired Edifier speaker path → air → Earthworks M50 → Audient microphone input 1 → the recording program. The operator reported approximately 10 cm separation. The exact acoustic reference plane, orientation and active cabinet assignment were not independently measured. Host outputs 1 and 2 received the same mono array; which physical speaker or speakers reproduced it was not fully logged.

Report scope. This is an offline review of existing recordings. No new playback, acquisition or hardware change was performed for this report. No neural or animal data were collected. Derived figures and records are publishable project artifacts; raw room audio remains local. The measured records are the evidence base, rather than the browser demonstration or manufacturer bandwidth claims [2, 3, 4, 5].

2 Dataset, acquisition and stimulus provenance

The principal dataset comprises two timing-probe captures and two SONIC E14 captures: **195.2 seconds** of native recording in four separately opened full-duplex streams. All four passed the saved transport checks; all recorded callback status flags were zero and no microphone sample reached the analysis clipping threshold of $|x| \geq 0.999$ full scale.

ID	Stimulus	Events	Time (s)	Frames	Mic peak (dBFS)
D1	Timing sweeps	20	11.6	1,113,600	-25.50
D2	Timing sweeps	20	11.6	1,113,600	-25.46
S1	E14 block	8,400	86.0	8,256,000	-27.65
S2	E14 block	8,400	86.0	8,256,000	-30.41

Table 2: Main captures at 96,000 samples/s. SONIC capture duration includes one second of silence before and after each 84-second stimulus block. Captured peak levels are digital dBFS, not dB SPL.

D1 `20261023-155401`
D2 `20261023-155500-roundtrip-repeat`
S1 `20261023-155615-sonic-e14-full`
S2 `20261023-160447-sonic-e14-full`

2.1 Acquisition configuration

The program used a named Audient iD14 device through Core Audio and `sounddevice` 0.5.6, with NumPy 2.3.5. Input channels 1/2 and output channels 1/2 were opened together at 96 kHz with float32 buffers, requested low latency and automatic block size. Timing runs used 908-frame callbacks. The exact source array, two-channel input array and per-buffer ADC/DAC timestamps were saved. The primary response is microphone input 1. Input 2 was *not* a validated electrical timing reference in these successful captures.

Input monitoring was kept attenuated in the working mixer setup. The microphone worked with the photographed switch configuration; phantom voltage, microphone gain, monitor attenuation, cabinet EQ and complete hardware routing were not numerically verified. No interface firmware update was performed, and its installed version was not recorded. These omissions limit exact physical reproduction.

2.2 Two distinct stimulus families

Timing probes (D1/D2): twenty 100 ms logarithmic sweeps from 100 Hz toward 20 kHz, with 20 ms raised-cosine end ramps, 500 ms onset spacing, and one-second padding. These locally specified probes are not SONIC stimuli.

SONIC reconstruction (S1/S2): 14 frequencies, in kHz,

$$\{2, 4, 6, 8, 9, 10.5, 17, 18.2, 19.5, 20.9, 22.4, 24, 25.8, 27\}.$$

Each 10 ms sine pip starts at zero phase and includes 2 ms ramps at both ends, with no silence between pips. Each block has 600 of every frequency: 8,400 pips and 84 seconds. A PCG64 shuffle with seed 20261023 generated the order; the same sequence was replayed in S2. Sources used −24 dBFS nominal digital peak. The frequency alphabet, pip duration, ramp duration and zero-phase convention follow the paper's Sheep E specification. The ramp shape, seed/order, exact counts, digital level and 96 kHz rate are declared local reconstruction choices, not original supplied SONIC audio [1, 3].

3 Timing analysis and results

3.1 The measured quantity

Let o be a source-array sample position and i the matching microphone-array position. The input/output array counters advance together in the native callback, but equal indices do not represent simultaneous ADC and DAC events. The estimated roundtrip is

$$\widehat{D}_{\text{app}} = \frac{i - o}{96{,}000}\,1000 \text{ ms}.$$

This sample-grid metric includes playback and recording buffering, interface conversion, speaker response, propagation and the microphone path. It is not a timed UI click, isolated speaker delay, first acoustic arrival or physical-only DAC-to-ADC latency. Reported driver latency and host timestamps were not added to or subtracted from it. PortAudio distinguishes requested/reported latency and backend-dependent timing behavior; metadata alone does not independently calibrate the physical chain [7, 2].

Each exact 100 ms probe was matched to the microphone with normalized waveform cross-correlation over a causal 0–100 ms lag search. Absolute correlation permits polarity inversion. The existing screen rejected weak, clipped, ambiguous or search-boundary matches and invalid transport. Local thresholds were $|r| \geq 0.35$, RMS above 10^{-6} full scale, and secondary-to-primary peak ratio below 0.95 after excluding the primary correlation lobe. These are analysis screens, not approved experimental acceptance limits.

Run	Accepted	Median (ms)	Range (ms)	$\lvert r\rvert$ range
D1	20/20	38.250	38.240–38.250	0.431–0.438
D2	20/20	38.375	38.365–38.375	0.433–0.438

Table 3: Sweep-based waveform roundtrip. The pooled median is 38.307 ms, with 0.135 ms total observed span. The two clusters correspond to separate stream starts.

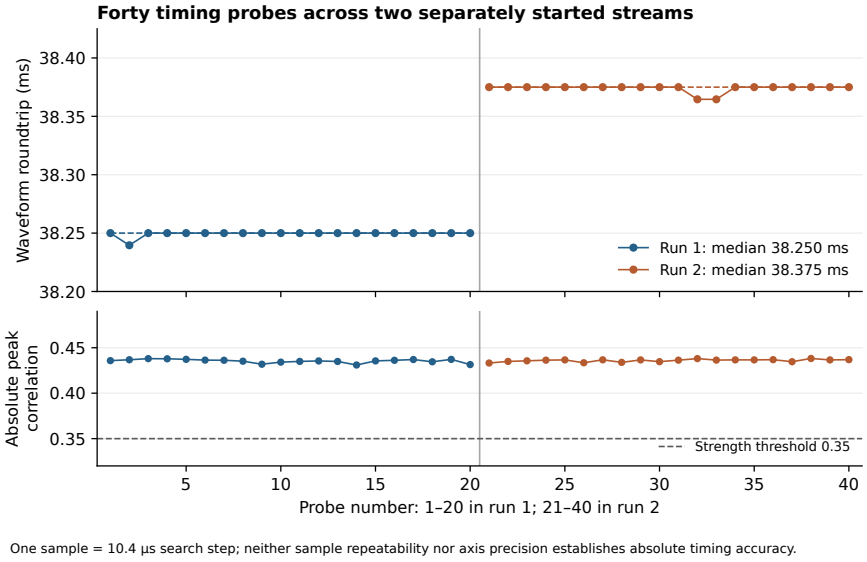


Figure 1: All 40 accepted integer-sample delay estimates and their correlation strength. Points are measured values; no confidence interval or calibrated accuracy bound is implied. Each run spans one sample in detected peak position.

3.2 Waveform evidence and timing interpretation

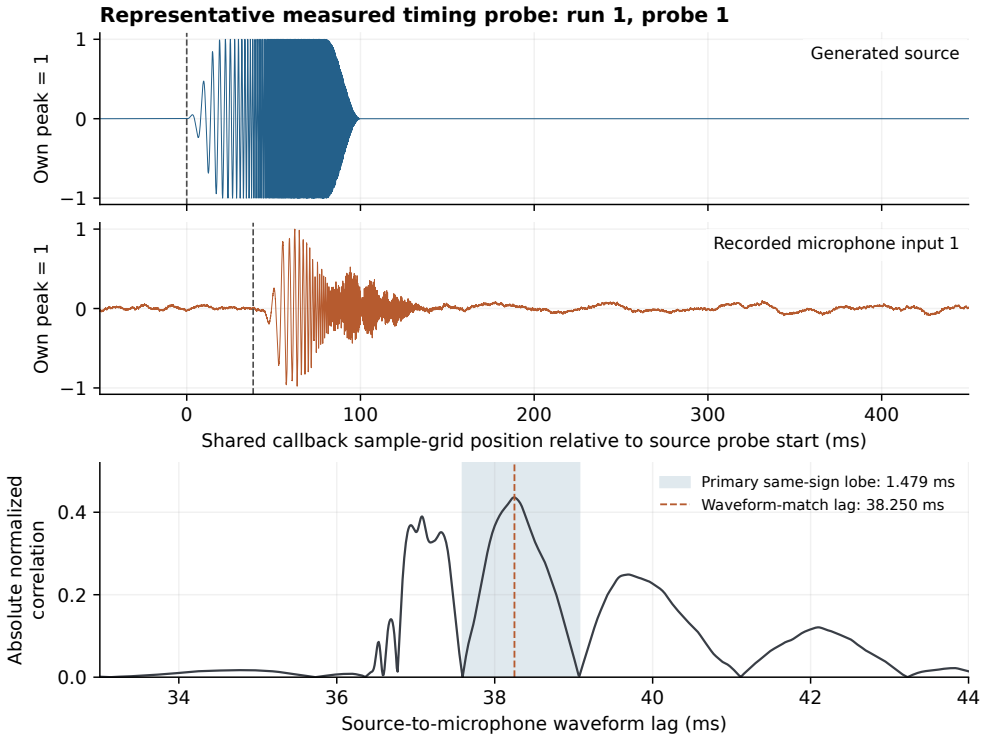


Figure 2: D1, first probe: actual source and microphone waveform in the application sample grid, followed by a zoom of the absolute normalized correlation. Trace amplitudes are normalized independently for display. The indicated correlation-lobe width is a shape diagnostic, not an uncertainty interval or first-arrival estimate.

Primary lobes spanned 141–142 samples (1.469–1.479 ms). Within-run peak positions varied by one sample (10.417 μs), whereas run medians shifted by 0.125 ms. Small sample-grid variation demonstrates repeatability of the estimator under this short test. It does not establish matching absolute accuracy: filtering, multipath, frequency-dependent phase and waveform choice can move the correlation maximum. Interpolated percentiles below one sample must not be reported as sub-sample timing accuracy.

Direct SONIC timing was not qualified. Five four-second windows from the recorded randomized tone sequence were tested separately. Four retained competing carrier peaks above the 0.95 ambiguity ratio; only one passed. The minimum accepted window screen was not met, so no SONIC-specific delay is reported. The 38.3 ms result is the separate nonperiodic-sweep estimate [6].

Cadence and latency are different. At 96 kHz, a 10 ms pip is 960 samples. A precomputed sequence can stream at that cadence while a pipeline has an approximately 38 ms roundtrip. Neither fact demonstrates accurate sub-millisecond acoustic pulses or synchronization to a neural recorder. Subtracting a fixed offset in software also does not remove physical latency.

4 SONIC tone pickup and frequency-dependent level

4.1 Analysis of the actual short-pip recordings

For every frequency in each block, a Hann-weighted complex projection measured the component at the commanded frequency in a 576-sample (6 ms) window. The window covers source-relative 2–8 ms after an approximate 38 ms content shift:

$$a_f = \frac{2\sum_n x[n]w[n]e^{-j2\pi fn/F_s}}{\sum_n w[n]}, \qquad A_{f,\mathrm{RMS}} = \frac{|a_f|}{\sqrt{2}}.$$

The projection is a descriptive component estimate, rather than an isolated steady-tone measurement. The 38 ms shift locates content; it is not a per-frequency delay result. Repeating extraction at 37 and 39 ms measures sensitivity to this local choice. Windowing reduces spectral leakage at the cost of finite resolution; it does not remove neighboring-tone or ringing effects [8, 3].

Tone (kHz)	S1 relative (dB)	S2 relative (dB)	S2–S1 (dB)	Min. contrast (dB)	Alignment span (dB)
2	0.00	0.00	+0.00	49.4	0.46
4	-2.70	-2.68	+0.01	52.1	0.56
6	-8.50	-8.52	-0.02	52.8	0.78
8	-7.36	-7.24	+0.12	55.9	0.61
9	-11.36	-11.79	-0.43	52.7	0.89
10.5	-8.66	-8.59	+0.07	58.1	0.85
17	-14.87	-14.75	+0.12	55.4	0.76
18.2	-18.62	-18.73	-0.12	52.2	0.82
19.5	-21.83	-22.14	-0.31	48.3	0.63
20.9	-40.36	-37.70	+2.66	31.9	5.90
22.4	-29.74	-30.22	-0.47	41.0	0.43
24	-36.80	-38.34	-1.54	10.5	0.62
25.8	-37.13	-35.44	+1.69	32.9	1.46
27	-28.59	-28.69	-0.09	39.1	0.54

Table 4: All-target medians, 600 pips/frequency/capture. Levels are relative to the same capture's 2-kHz median. Minimum contrast is the lowest pre-block or other-tone contrast across S1/S2, in amplitude-ratio dB, not calibrated SNR. Alignment span is the larger 37–39-ms sensitivity span across the two captures.

All 14 groups contained components above both matched-frequency pre-block and in-block controls in both captures. Other-tone controls excluded the target and near first-through-sixth harmonic relationships (334 Hz tolerance) at the current event and two neighbors on either side. A second target analysis removed nearby same-frequency/near-harmonic events, retaining 159–454 pips per group. Its median changes were at most 0.224 dB. This includes excluding 10.5 kHz neighbors when examining 20.9 kHz, because their second harmonic is near 21 kHz.

The weakest conservative contrast was 10.51 dB at 24 kHz; 20.9 kHz was most alignment-sensitive (5.90 dB). These observations warrant particular caution at the weak high-frequency groups. Per-pip 5th–95th percentiles are descriptive amplitude spreads, not confidence intervals. The repeated seeded order also means the 1,200 observations per frequency are not 1,200 independent randomized sessions.

4.2 Measured relative-response comparison

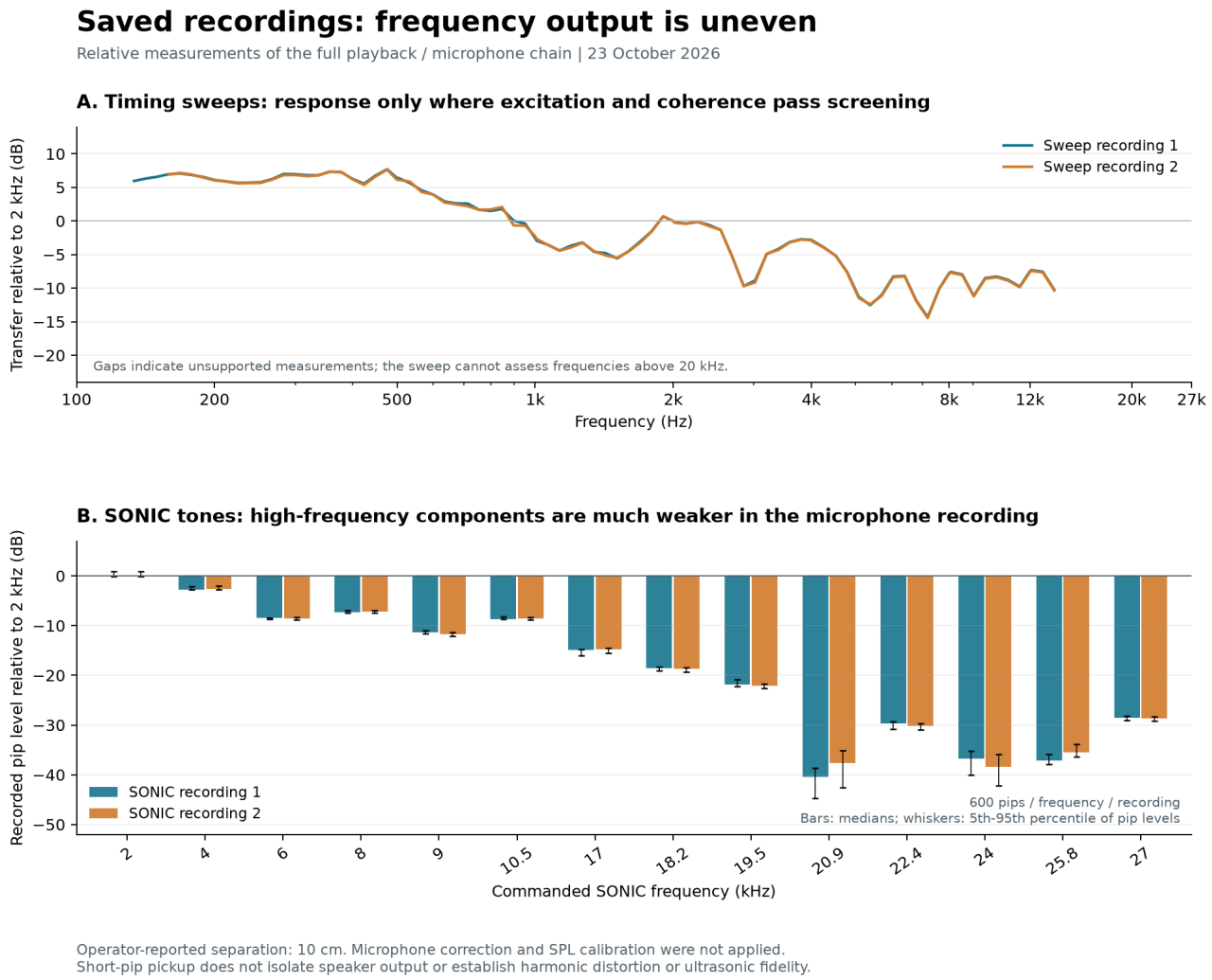


Figure 3: Two independent analyses of existing recordings. Top: qualified sweep H1 magnitude bands, calculated separately for D1 and D2. Gaps are unsupported measurements. Bottom: SONIC pip component medians relative to 2 kHz; whiskers show the within-group 5th–95th percentile spread. These are full-chain, uncorrected measurements at the saved position.

The main reproducible finding is **uneven recorded level**. Relative to 2 kHz, the 2–10.5 kHz groups span about 0–12 dB attenuation; 17–19.5 kHz are about 15–22 dB lower; 20.9–25.8 kHz about 30–40 dB lower; and 27 kHz about 29 dB lower. The weak groups show more run/alignment sensitivity. Equal digital peaks therefore cannot be treated as equal acoustic presentation levels.

At matching frequencies and times, input-1 component medians exceeded input-2 values by approximately 57–97 dB, and input 2 showed no corresponding event-specific rise over its pre-block floor. This is supporting comparator evidence, not proof of acoustic origin: input gains/wiring differ, and no speaker-disconnected isolation capture was made. The results do not isolate speaker response from room reflections, microphone response, routing or geometry.

5 Relative response from the timing sweeps

For each run independently, 0.5-second rectangular windows from −0.1 to +0.4 s relative to each probe onset include the source sweep and delayed microphone response in the same flat window. A full Hann window was avoided because delay would move the source and response onto different amplitude weights. With averages over the 20 repeated probes,

$$H_1(f) = \frac{\langle Y(f)X^*(f)\rangle}{\langle |X(f)|^2\rangle}, \qquad \gamma^2(f) = \frac{|\langle Y(f)X^*(f)\rangle|^2}{\langle |X(f)|^2\rangle\langle |Y(f)|^2\rangle}.$$

Source-power-weighted RMS magnitudes and magnitude-squared coherence were summarized in 1/12-octave bands; levels were normalized to the qualified 2 kHz band. Coherent response was compared with a source-silent 0.1–0.6 s baseline. The local masks require coherence ≥ 0.90, source mean-bin power within 30 dB of the strongest comparable band, response at least 10 dB above baseline, and valid unclipped transport. Pip phase-concentration diagnostics are different from this cross-spectral coherence and should not be interpreted interchangeably.

kHz	D1 (dB)	D2 (dB)	γ_1^2	γ_2^2	Screen
0.1	–	–	0.425	0.321	Masked
0.2	6.10	6.05	0.998	0.998	Pass
0.5	6.40	6.15	0.991	0.992	Pass
1	-2.53	-2.26	0.975	0.968	Pass
2	0.00	0.00	0.996	0.997	Pass
4	-2.81	-2.87	0.993	0.994	Pass
8	-7.75	-7.85	0.980	0.977	Pass
12	-7.50	-7.62	0.950	0.949	Pass
16	–	–	0.671	0.621	Masked
18	–	–	0.115	0.114	Masked
20	–	–	0.046	0.052	Masked

Table 5: Representative 1/12-octave H1 band summaries relative to 2 kHz. A dash means masked, not zero gain. Coherence alone is insufficient; excitation and baseline conditions must pass as well.

84 of 101 tested centers passed in both runs, from 158.74 Hz through 13561.13 Hz. The qualified combined magnitudes span approximately 22.05 dB. No physical bandwidth cutoff is inferred from those endpoints. The sweep's final 20 ms fade covers roughly 6.93–20 kHz; near 20 kHz source power is extremely weak. Naive division there would create misleading high-frequency gain. The 100 Hz and 16/18/20 kHz anchors are masked.

Complex spectra were not averaged across the two stream starts: their 0.125 ms offset would itself create phase cancellation. Each run has its own H_1; only qualified magnitude summaries are compared or pooled. Finite window length, synchronous background noise, time-varying conditions and response outside the window remain possible biases. This is a linear relative-response screen at one digital peak, not a THD, compression or speaker-only calibration [4].

6 SONIC sequence example and experimental meaning

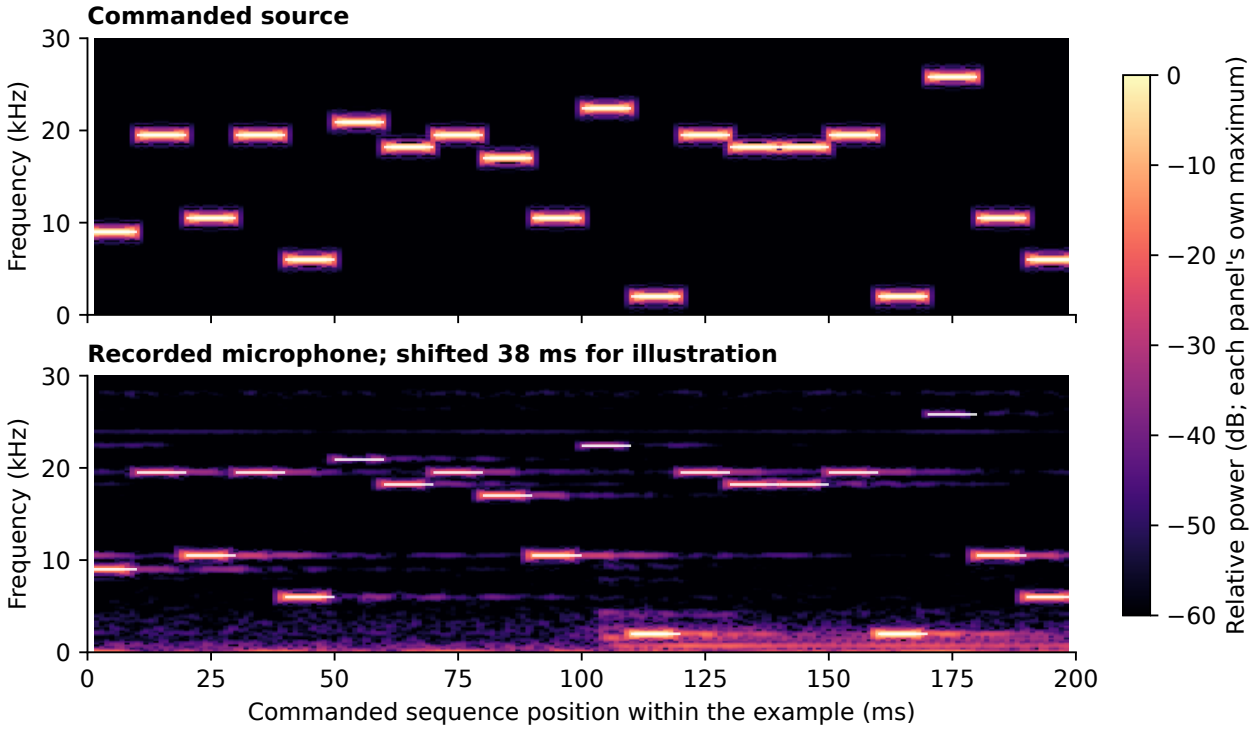


Figure 4: Actual 200-ms excerpt from S1. Commanded frequencies are shown with a microphone time–frequency display. The microphone display is shifted by approximately 38 ms to locate content, not to establish SONIC latency. The 4-ms Hann window has a 250-Hz inverse-window scale and a broader mainlobe; the hop is 1 ms. The view does not resolve exact acoustic onset or certify every high-frequency pip.

6.1 What carries over from the paper

The local reconstruction reproduces Sheep E's listed frequency alphabet, 10 ms pip duration, zero starting phase, 2 ms end-ramp duration and consecutive presentation. The paper used precomputed waveforms at 192 kHz through an NI USB-6361, TDT SA1 and MF1, reporting approximately 80 dB at sheep-ear distance. It describes 8,000–15,000 tones per block and at least 12 blocks per session. Our two captures replay one locally seeded 8,400-pip block at 96 kHz on different hardware; original order, ramp shape and digital level were not available. The paper's ear-position level is not an established or approved local output level [1].

The paper's 56/11 ms intrinsic neural-interface delays have a different definition: they exclude auditory physiology and computation, and the 11 ms result is a narrower session/window sweep. Comparing them directly with this 38.3 ms audio application roundtrip would be invalid. No neural decoding, physiological latency or ITR can be inferred from these bench recordings. Published SONIC animals were awake and alert; planned Rice work in sedated sheep is a separate adaptation whose responses remain unmeasured [1, 10].

7 Limits and prioritized next bench tests

The evidence establishes operation, short-run roundtrip repeatability and frequency-dependent recorded pickup. It leaves **absolute SPL, isolated speaker fidelity, distortion, physical onset and neural-clock alignment open**. The M50 has a serial-specific calibration file; none was applied here. Nominal manufacturer bandwidth or sensitivity cannot replace calibration of the delivered microphone, preamp gain and complete acoustic geometry [9].

The 96 kHz stream has a 48 kHz Nyquist frequency, so a 27 kHz fundamental is representable in principle. Its 54 kHz second harmonic is not. At 24 kHz the second harmonic lies at Nyquist; higher tones have second harmonics above it. Faster conversion alone would still require calibrated microphone and analog front-end coverage. Short continuous pips also mix source-window leakage, neighboring tones and ringing, so these recordings do not provide full THD or THD+N. No numeric flatness, level, distortion or onset tolerance has been approved.

Order	Proposed next test	Evidence required before closing the question
1	Fix the physical configuration	Record the active cabinet/channel, reference plane, tip position and angle, room/background, gains, EQ, routing, sample rate and firmware version. Use the intended measurement position.
2	Calibrate the measurement chain	Load the serial-matched M50 correction and establish level conversion with a suitable acoustic reference at the recorded gain. Target SPL and convention: TBD with Dr. Luan.
3	Isolate acoustic pickup	Compare playback with the speaker path active and silent/disconnected while retaining interface/microphone connections. Quantify residual electrical feedthrough.
4	Measure each frequency separately	Use settled, separated tones across all 14 frequencies, repeated orders and documented digital levels. Assess response, noise, ringing and measurable harmonic orders. Level/flatness/distortion tolerances: TBD.
5	Validate independent timing	Establish a known electrical return alongside the microphone, verify channel skew, and repeat stream starts, buffer settings and longer captures. Timing accuracy/jitter requirement: TBD.
6	Integrate the neural timebase	Validate the eventual trigger/clock mapping to the neural recorder. Confirm the animal-position acoustic stimulus before interpreting neural response windows. No animal work is established by this report.

Table 6: Proposed validation sequence, not tests already performed. Suggested bench owner: Howard; experimental targets and acceptance criteria: TBD with Dr. Luan. No purchase, exposure level, procedure or animal protocol is approved here.

The immediate decision is to retain the working setup for controlled bench characterization, while treating full-band level uniformity and calibrated timing as open measurements. The present high-frequency attenuation does not identify a defective component: speaker directivity, two-cabinet interference, position, room response, hardware settings and microphone correction remain hypotheses to test.

8 Reproducibility and evidence ledger

The numeric tables are generated from the saved verification records, rather than manually entered. A fresh offline audit reproduced both timing/H1 analyses and both pip analyses and matched all 34 checked SHA-256 references. Figures use actual saved samples or measured aggregate values; synthetic signals are used only in software tests. Original raw arrays, timestamps and manifests remain local under ignored `.work/latency/runs/`.

Provenance files. `data-manifest.json` records the verification-file, code and pinned-paper hashes. `images/figures-manifest.json` records exact source windows, normalization, STFT settings and raw/figure hashes. These accompany the editable LaTeX project. The pinned paper stays unchanged with its version, checksum and CC BY-NC-ND 4.0 notice; all plotted experimental figures are local data.

Historical records retain their original scope. Earlier silent/weak microphone checks and missing electrical-return analyses did not establish a delay. The later source-array roundtrip does not retroactively validate input 2. D2's initial manifest omitted a native-acquisition flag; the original was retained as `manifest.as-acquired.json`, and the documented metadata amendment added that flag without changing audio or timestamps [2].

Offline reproduction (requires the private saved captures):

```
python -m latency.roundtrip --analyze-run RUN_FOLDER
python -m latency.pip_response
python -m latency.sweep_response
python scripts/create_frequency_response_figure.py
python scripts/create_full_bench_report_data.py
python scripts/create_full_bench_report_figures.py
```

These analysis/figure commands do not play sound. Compile this project's `main.tex` to regenerate the report. The dataset is small, covers two stream starts per stimulus family and repeats the same source sequences; no long-term or independent-session confidence interval is claimed.

References

[1] *SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces.* bioRxiv v1, posted 2 October 2025. Methods 6.1–6.2 (p. 11), animal state 6.6 (p. 13), intrinsic-delay results (pp. 5, 7–8). https://doi.org/10.1101/2025.09.30.679683.

[2] Project measured timing record, 23 October 2026. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/docs/verification/2026-10-23_Audient_Playback_Microphone_Delay.json.

[3] Project two-capture E14 pip-frequency review, 23 October 2026. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/docs/verification/2026-10-23_SONIC_E14_Pip_Frequency_Pickup.json.

[4] Project screened sweep H1 response, 23 October 2026. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/docs/verification/2026-10-23_Sweep_Relative_Response.json.

[5] Project basic playback/recording verification. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/docs/verification/2026-10-23_Audient_Basic_Playback_Recording.json.

[6] Project E14 playback and unqualified direct-tone timing review. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/docs/verification/2026-10-23_SONIC_E14_Playback.json.

[7] PortAudio, *Buffering, Latency and Timing Implementation Guidelines* (draft guidance), accessed 23 October 2026. https://github.com/PortAudio/portaudio/wiki/BufferingLatencyAndTimingImplementationGuidelines.

[8] NI, *Understanding FFTs and Windowing*, accessed 23 October 2026. https://www.ni.com/en/shop/data-acquisition/measurement-fundamentals/analog-fundamentals/understanding-ffts-and-windowing.html.

[9] Earthworks Audio, M50 specifications and individual electronic calibration file, accessed 23 October 2026. https://earthworksaudio.com/measurement-microphones/m50/.

[10] Project working directions: planned Rice sedated-sheep adaptation. https://github.com/HowardWHSrun/audiotory_benchmark/blob/main/AGENTS.md.